Chapter 46 Magnetic Forces
Calculus Physics II

<u>Force on a moving point charge ($\vec{F}_M$) in a magnetic field ($\vec{B}$)</u> - For a particle to experience a force due to a magnetic field it must meet three conditions.

1) The object must have a net charge.
2) The object must be moving.
3) The object must have a component of its velocity vector perpendicular to the magnetic field lines.

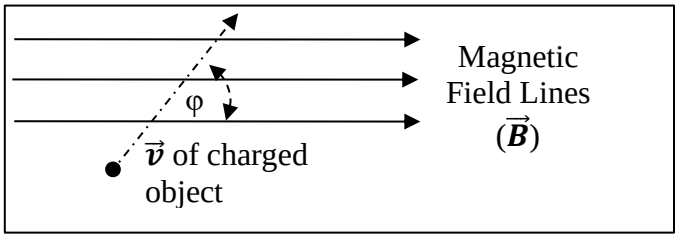
The magnetic force vector $\vec{\mathbf{F}}_\mathbf{M}$ is equal to the cross product of the velocity vector $\vec{\mathbf{v}}$ of the charged object with the external magnetic field vector $\vec{\mathbf{B}}$ times the net charge q of the object.

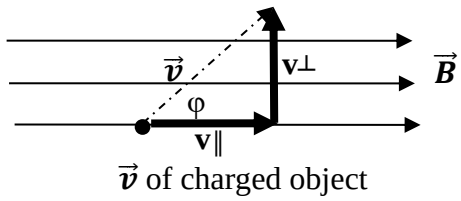
$$\vec{\mathbf{F}}_\mathbf{M} = \left(\vec{\mathbf{v}}\mathbf{x}\vec{\mathbf{B}}\right)q \qquad \text{Units are Newton's (N)}$$

$$\vec{\mathbf{F}}_\mathbf{M} = [|\vec{\mathbf{v}}||\vec{\mathbf{B}}|\mathrm{q}\sin(\varphi)]\hat{\mathbf{F}}$$

$$\left|\vec{\mathbf{F}}_\mathbf{M}\right| = |\vec{\mathbf{v}}||\vec{\mathbf{B}}|\mathrm{q}\sin(\varphi) = \mathrm{q}\mathbf{v}_\perp\mathbf{B} \qquad \text{where} \quad \mathbf{v}_\perp = |\vec{\mathbf{v}}|\sin\varphi$$

$\vec{\mathbf{F}}_\mathrm{M}$ – Force due to magnetic field.
q – net charge.
$\vec{v}$ – velocity of charge.
$\vec{B}$ – magnetic field strength.
φ – the angle between the velocity vector and the magnetic field vector.
$\mathbf{v}_\perp$ – velocity component perpendicular to magnetic field lines.
$\hat{\mathbf{F}}$ – unit vector in the direction of the magnetic force and determined by the right-hand rule # 1.





- An object must have a net charge (q) in order to experience a force in a magnetic field.
- The force on a moving positive charged object is equal in magnitude but in the opposite direction of that on a moving negatively charged object.
- Charged objects that are not moving (v = 0 m/s) do not experience a magnetic force.
- The charged object experiences no force moving in a magnetic field when it travels along a magnetic field line ($\varphi = 0°$).

- The maximum force the charged object feels when its velocity direction is perpendicular to the magnetic field lines (φ = 90°).
- The unit vector $\hat{\mathbf{F}}$ is in the direction of the Force vector $\vec{\mathbf{F}}$ and is determined by the Right-hand rule for cross products. $(\vec{\mathbf{v}}\mathbf{x}\vec{\mathbf{B}})$
- The direction of the force is perpendicular to both the velocity and magnetic field vector.

The Magnetic Field at a point in space is represented by a vector $\vec{B}$ and a charged object that is moving with an angle relative the magnetic field lines will experience a force $\vec{F}_M$ so that the magnetic field strength is given by

$$|\vec{B}| = \frac{|\vec{\mathbf{F}}_\mathbf{M}|}{q|\vec{v}|\sin\varphi} = \frac{\boldsymbol{Force}}{\boldsymbol{unit\ moving\ charge}}$$

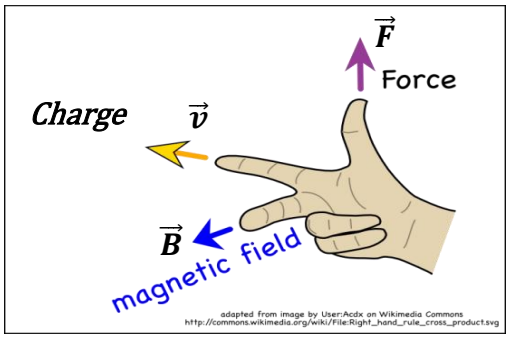
Drawing the Magnetic Field – Representing the Magnetic Field on paper.

Field lines	Field Out of page	Field into Page
→	.....	x x x x
→	.....	x x x x
→	.....	x x x x
→	.....	x x x x

Direction of the magnetic force on a moving charge – Right hand rule # 1 (RHR) – for a ***positive charge!***

1) Point with the index finger in the direction of the velocity vector (1st vector) of the charged particle or current,
2) with your next 3 fingers point in the direction of the magnetic field vector (2nd vector) and
3) your thumb points in the direction of the force vector.
4) If the charge is negative the force is in the opposite direction as the positive charge.

Caution - make sure you use your right hand!!!



Finger placement for the right-hand rule: Note that the Force vector (thumb) is always perpendicular to the velocity and magnetic field vectors.

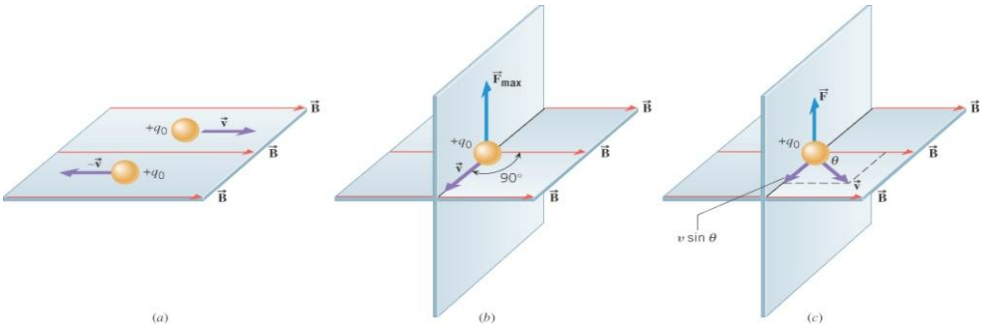
$$\vec{\mathbf{F}}_\mathbf{M} = (\vec{\mathbf{v}}\mathbf{x}\vec{\mathbf{B}})q$$

$$|\vec{\mathbf{F}}_\mathbf{M}| = |\vec{\mathbf{v}}||\vec{\mathbf{B}}|q\sin(\varphi)$$

The magnitude of the magnetic force depends on

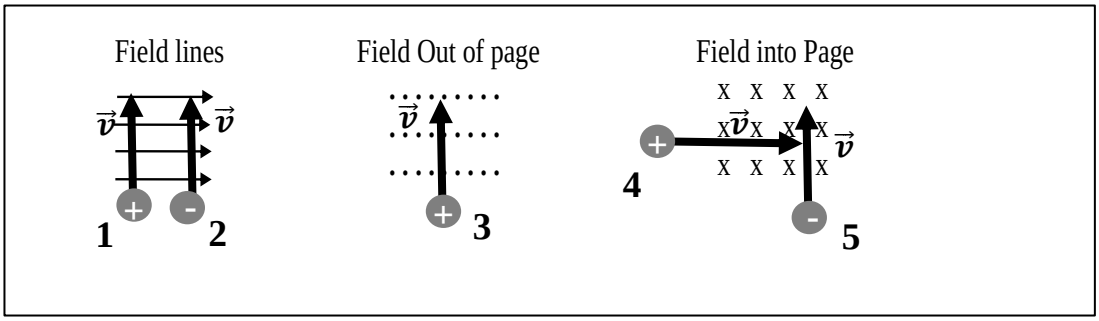
1) Magnitude of net charge.
2) Velocity of charged object perpendicular to field ($\vec{v}\sin\varphi$).
3) Magnitude of magnetic field.

- The force on a moving point charge in a magnetic field is always perpendicular to path of the charge.
- The direction of the magnetic forces is perpendicular to both the magnetic field and the velocity, or the force is perpendicular to the plane of $\vec{B}$ and $\vec{v}$.



(a) Velocity direction of the charged particle is along the field line – There is NO magnetic force created.
(b) Veloicty direction of the charged particle is perpendicular to the magnetic field – The magnetic force is maximum.
(c) The velocity direction of the magnetic field is at an angle (φ) to the magnetic field line the magnetic force will be reduced by "sin(φ)".

Examples of the right-hand rule # 1 – Using the field and velocity directions of the charged point charge below determine the direction of the force on the charged sphere using the right hand rule.



Particle 1 - Point your index finger in the direction of the velocity, which is towards the top of the page, then rotate your hand so that the other three fingers are along the magnetic field, which is to the right of the page, the direction of your thumb is the direction on the force on the particle, which is into the page.

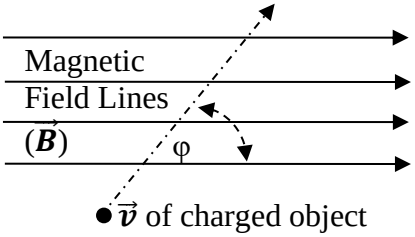
Particle 2 – Repeat process above, because the charge on the sphere is negative the force will be pointing in the opposite direction as the positive charge, the force points out of the page.

Particle 3 - Point your index finger in the direction of the velocity, which is towards the top of the page, then rotate your hand so that the other three fingers are along the magnetic field, which is pointing out of the page, the direction of your thumb is the direction of the force on the particle, which is to the right.

Particle 4 - Point your index finger in the direction of the velocity, which is towards the right side of the page, then rotate your hand so that the other three fingers are along the magnetic field, which is pointing into the page, the direction of your thumb is the direction on the force on the particle, which is towards the top of the page.
Particle 5 - Point your index finger in the direction of the velocity which is towards the top of the page, then rotate your hand so that the other three fingers are along the magnetic field which is pointing into the page, the direction of your thumb is the direction on the force on a positive particle is to the left therefore a negative particle would experience a force to the right.

Example: A charged particle (q = 200 μC) is moving in a magnetic field with a strength of $|\vec{\mathbf{B}}| = 2 \times 10^{-3}$ T and a velocity of $|\vec{\mathbf{v}}| = 6 \times 10^{6}$ m/s.

A) If the velocity makes an angle of 90° to the magnetic field line what is the magnitude of the magnetic force the particle experience and in what direction?



$F_M = qvB\sin\varphi$

$= (200 \times 10^{-6}\ C)(6 \times 10^{6}\ m/s)(2 \times 10^{-3}\ T)\sin(90°) = 2.4\ N$

B) If the velocity makes an angle of 90° to the magnetic field line what is the magnitude of the magnetic force the particle experience and in what direction?

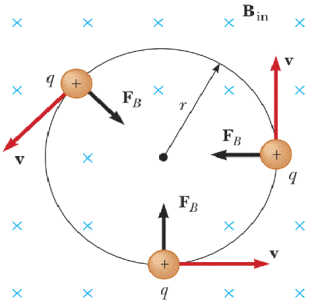
$F_M = qvB\sin\varphi$

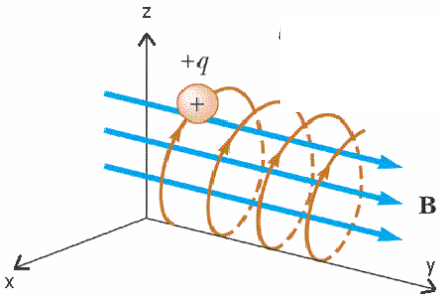
$= (200 \times 10^{-6}\ C)(6 \times 10^{6}\ m/s)(2 \times 10^{-3}\ T)\sin(30°) = 1.2\ N$

C) If the velocity makes an angle of 0° to the magnetic field line, the particle is moving parallel to the lines, what is the magnitude of the magnetic force the particle experience and in what direction?

$F_M = qvB\sin\varphi = (200 \times 10^{-6}\ C)(6 \times 10^{6}\ m/s)(2 \times 10^{-3}\ T)\sin(0°) = 0\ N$

Centripetal Acceleration and the Magnetic Field - In a magnetic field the force on the moving charged particle remains perpendicular to the velocity and is directed towards the center of the circular path. Because the magnetic force is always in the direction towards the center, the charged particle experiences a centripetal acceleration and travels in a circular path in a uniform magnetic field.





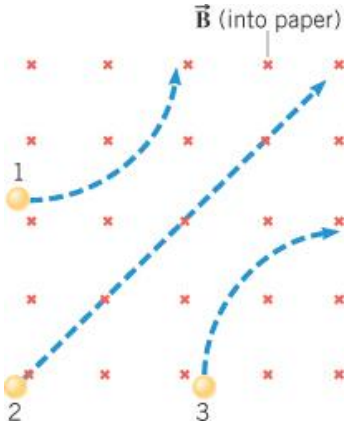
Example: What is the charge on each particle 1, 2 and 3 shown below?

Using the right-hand rule # 1 determines the direction of the force on the charge particle and the type of charge that would result in the motion shown.

1) The force direction from the right-hand rule on particle 1 is towards the center of the circle therefore it must be positive because it is in the correct direction for this circular motion.

2) There is no force on this particle therefore it must not have a net charge, it is neutral.

3) From the right-hand rule # 1 the force direction points away from the center therefore it would not have this curvature if it was positive, it must be a negatively charged particle for the force to point towards the center.



Path of charged object moving in Magnetic Field - Because the Magnetic force is always perpendicular to the direction of motion, it will travel in a circular path. If it is traveling in a circular path then there is a centripetal force acting on the particle, and it must be equal to the Magnetic Force that the object is experiencing. The centripetal acceleration depends on the velocity of the particle and the radius of the path ($|\vec{a}_c| = v^2/r$)

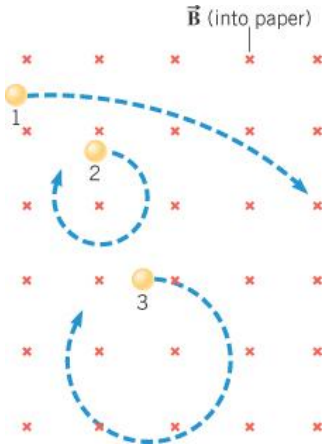
$$\Sigma\vec{F} = m\vec{a}_c \quad \text{the magntude of the force } \left|\vec{F}_M\right| = m|\vec{a}_c|$$

$$qvB\sin\varphi = mv^2/r \quad \text{solving for r}$$

$$r = \frac{mv}{qB\sin\varphi}$$

r is the radius of charged particles path in $\vec{B}$ field

Example: Three charged particles move in a uniform magnetic field



1. If three charged particles (same charge) are moving at the same velocity in a uniform magnetic field, then the particles with the larger radius have the larger mass.

$$m_1 > m_2 > m_2$$

2. If the mass and charge of the particles are the same, then the particles with the larger velocities have the larger radius.

$$v_1 > v_2 > v_2$$

3. If the charged particles have the same mass and velocity but different charge, then the particles with the larger charge have the smaller radius.

$$q_1 < q_2 < q_2$$

Example: A proton and electron are moving perpendicular ($\varphi = 90°$) to magnetic field ($|\vec{B}| = 1$ T) at a velocity of 3×10^6 m/s, what is the radius of their circular motion in the magnetic field?

Electron: $e^- = 1.6 \times 10^{-19}$ C, negative charge, $m_e = 9.1 \times 10^{-31}$ kg

Proton: $e^+ = 1.6 \times 10^{-19}$ C, positive charge, $m_p = 1.67 \times 10^{-27}$ kg

For the Proton

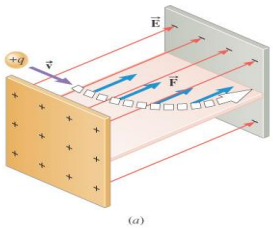
$$r = \frac{mv}{qB\sin\varphi} = \frac{(1.67x10^{-27}kg)(3x10^6m/s)}{(1.6x10^{-19}C)(1T)} = 0.03\,m$$

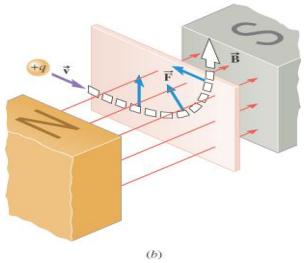
For the electron

$$r = \frac{mv}{qB\sin\varphi} = \frac{(9.11x10^{-31}kg)(3x10^6m/s)}{(1.6x10^{-19}C)(1T)} = 1.7x10^{-5}\,m$$

- Because the proton is almost 2,000 time more massive than the electron the radius of the circle it makes in the magnetic field is much larger than the electron.

Comparison of Magnetic and Electric Forces and Fields





- In the Magnetic field, the force a moving charged particle experiences is always perpendicular to the magnetic field lines.
- In Electric fields the charged particle experiences a force along the field lines.
- Magnetic Fields have no beginning or end but are loops.
- Electric Fields begin on a positive charge and end on negative charge or at infinity.
- Electric fields produce a translational acceleration along the field electric field lines.
- Magnetic fields create a centripetal acceleration.
- Because the magnetic force is perpendicular (90°) to the direction of motion of the charged particle, there is no work done on the object by the Magnetic Field.
- Electric field can do work on a charged particle.

Lorentz Force - The Lorentz force describes the total electromagnetic force on a charged particle, combining the effects of both electric and magnetic fields. It is given by the vector equation.

$$\vec{F} = \underbrace{q\vec{E}}_{\text{Electric force}} + \underbrace{q\vec{v}\,x\,\vec{B}}_{\text{Magnetic force}}$$

The electric force component acts parallel to the electric field, while the magnetic force component is perpendicular to both the velocity and the magnetic field.

- The electric field component acts on the charged particle due to the presence of an electric field, it acts in the direction of the field and is independent of the velocity of the charged particle.
- The magnetic field component acts on a charged particle when it moves through a magnetic field perpendicular to both the velocity and magnetic field lines. The magnetic force is dependent on the velocity of the charged particle.

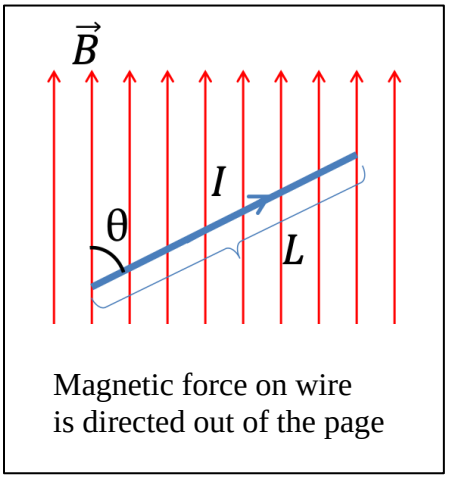
Force on a current in a magnetic field - Since a current is simply a stream of charged particles, a current moving through a wire experiences a force due to an external magnetic field. For a straight wire of length L in a uniform magnetic field the magnetic force is give by

$$\vec{\mathbf{F}}_{\mathbf{M}} = q(\vec{\mathbf{v}}\mathbf{x}\vec{\mathbf{B}}) = q\left[\left(\frac{\Delta\vec{\mathbf{r}}}{\Delta t}\right)\mathbf{x}\vec{\mathbf{B}}\right] \text{ but } \frac{\Delta q}{\Delta t} = I \text{ and } \Delta\vec{r} = \Delta\vec{L}$$

$$\vec{\mathbf{F}}_{\mathbf{M}} = \left[I\vec{L}\left|\vec{\mathbf{B}}\right|\sin(\varphi)\right]\hat{F}$$

$$\boxed{\left|\vec{\mathbf{F}}_{\mathbf{M}}\right| = I\left|\vec{L}\right|\left|\vec{\mathbf{B}}\right|\sin(\varphi)}$$

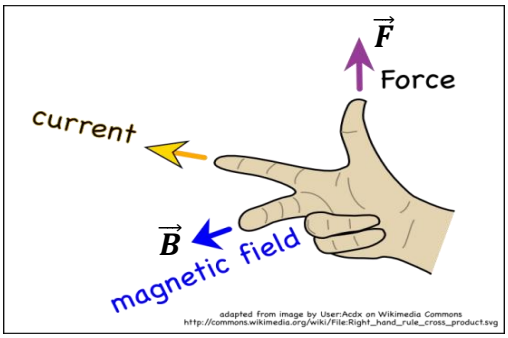
$\vec{\mathbf{F}}_{\mathbf{M}}$ - force due to a current in a magnetic field (N).
$\hat{\mathbf{F}}$ - the unit vector in the direction of the magnetic force.
I – current in wire (A) for a positive charge.
L – length of wire in magnetic field (m).
$\vec{\mathbf{B}}$ – magnitude of magnetic field (T).
φ – angle between wire and magnetic field (°).

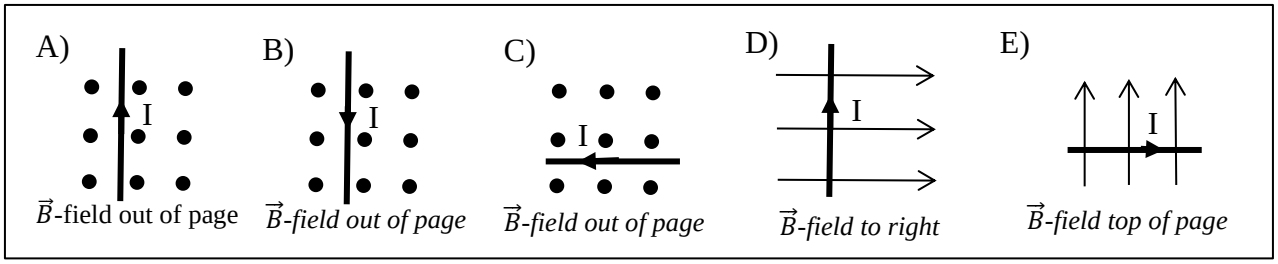


Magnetic force on wire
is directed out of the page

Examples of right-hand rule: If a wire is placed in a magnetic field with the current direction as shown, what direction is the magnetic force on the wire?

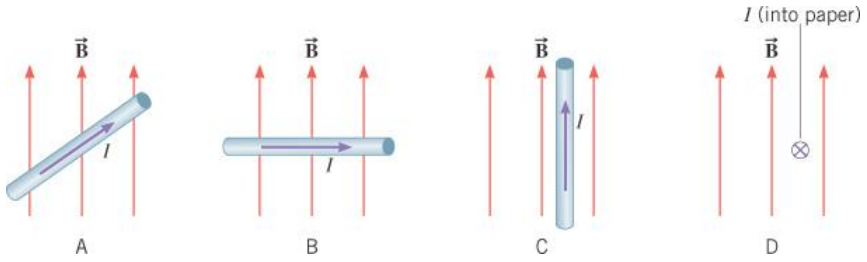
The direction is obtained by using the right-hand rule # 1 using the current direction as the velocity direction of the charge. Pointing the index finger in the direction of the current and rotating the hand so that the other three fingers line up with the magnetic field, the direction of the thumb is the direction of the magnetic force.





A) The magnetic force due to the current will push the wire to the right.
B) The magnetic force due to the current will push the wire to the left
C) The magnetic force due to the current will push the wire toward the top of the page
D) The magnetic force due to the current will push the wire into the page.
E) The magnetic force due to the current will push the wire out of the page.

Example: In the diagram below assume that a L = 1 meter wire is placed in a magnetic field ($|\vec{B}|$ = 2 x 10^{-3} T) as shown above, all the wires have the same current of I = 2 A flowing through them.



A) What is the magnitude and direction of the magnetic force on the wire due to the current in the magnetic field if the angle the wire makes with the field line is 45°?

$|\vec{F}_M|$ = ILBsinφ = (2 A)(1 m)(2 x 10^{-3}T)sin(45°) = 2.8 x 10^{-3} N

The force on the wire is out of the page (from RHR #1)

B) What is the magnitude and direction of the magnetic force on the wire due to the current in the magnetic field if the angle the wire makes with the field line is 90°, the wire is perpendicular to the field line?

$|\vec{F}_M|$ = ILBsinφ = (2 A)(1 m)(2 x 10^{-3}T)sin(90°) = 4.0 x 10^{-3} N

The force on the wire is out of the page (from RHR #1)

C) What is the magnitude and direction of the magnetic force on the wire due to the current in the magnetic field if the angle the wire makes with the field line is 0°, the wire is along a field line?

$|\vec{F}_M|$ = ILBsinφ = (2 A)(1 m)(2 x 10^{-3}T)sin(0°) = 0 N

There is no magnetic force on the wire

D) What is the magnitude and direction of the magnetic force on the wire due to the current in the magnetic field if the angle the wire makes with the field line is 90° into the page?

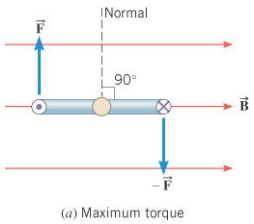
$|\vec{F}_M|$ = ILBsinφ = (2 A)(1 m)(2 x 10^{-3}T)sin(90°) = 2.8 x 10^{-3} N

The force on the wire is to the right (from RHR #1)

Torque on a Current Carrying Coil - When a current carrying loop of wire is placed in a magnetic field, it experiences a force ((a) in diagram, φ = 90°), the force tends to rotate the loop such that its normal (a line perpendicular to the plane of the loop) becomes aligned with the field ((b) in diagram, φ = 0°). This force of rotation is a torque given by Torque = Fl where F is the applied force and l is the lever arm but l, where φ is the angle between the radius and the force. The Torque ($\vec{\tau}$) on a current carrying loop is given by a cross product.

$$\vec{\tau} = \vec{r}x\vec{F}_M$$

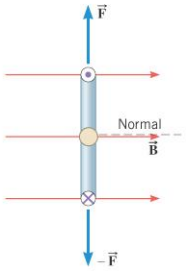
$$|\vec{\tau}| = NIABsin(\varphi) \qquad \text{Units of Nm}$$



(a) Maximum torque

where

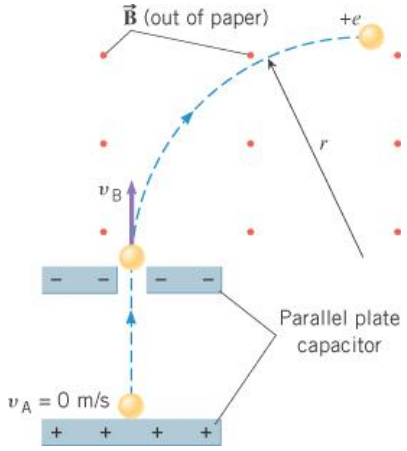
N - is the number of turns or loops of wire.
I - is the current flowing in the wire.
A - is the cross-sectional area of loop.
B - is magnetic field strength.
φ - is the angle between normal to the plane of the loop and the magnetic field.



(b) Zero torque

Applications for Magnetic forces and fields

Mass Spectrometer – uses the fact that the radius of the circle of a charged particle depends on its charge and mass, the charged particle first goes through a potential difference (ΔV) to give it an initial velocity ($\vec{v}$) then it goes into a known magnetic field (pointed out of the page) where the radius ($\vec{r}$) is measured, and the mass can be determined.



Using the forced on a charged particle in circular motion, the magnetic force is the centripetal force, therefore.

$$\left|\vec{F}_M\right| = m|\vec{a}_c|$$

$$qvBsin\varphi = mv^2/r$$

Solving for m, $m = rqvB/v^2$ assuming $\varphi = 90°$

Using conservation of energy $\Delta KE + \Delta PE_E = 0$

therefore, $\frac{1}{2}\, mv^2 = q\Delta V$

Solving for v, $v^2 = 2q\Delta V/m$ substituting this into above equation for mass,
Mass of charged particle with charge q and make a radius r in a magnetic field B that goes through a potential ΔV.

$$m = \frac{qr^2B^2}{2\Delta V}$$

The mass of a charged object can be determined by knowing the net charge, magnitude of the magnetic field, potential the charged particle was accelerated through and the radius of the path in the magnetic field.

Example: If a mass spectrometer uses a particle accelerator with a voltage difference of 1,000 V and a magnetic field strength of 2 T

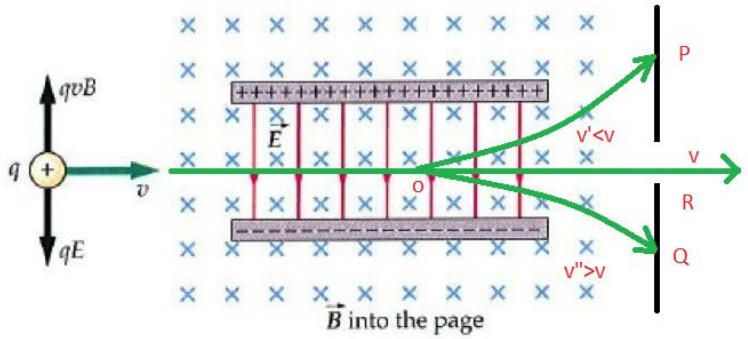
Determine the mass of a particle that has a charge of 1.6 x 10^{-19} C and produces a radius of 0.0079 m in the magnetic field?

$$m = \frac{qr^2B^2}{2\Delta V} = \frac{(1.6\ x\ 10^{-19}C)(0.0079\ m)^2(2T)^2}{2(1{,}000V)} = 2\ x\ 10^{-26}kg$$ This is a carbon atom!

Determine the mass of a particle that has a charge of 1.6 x 10^{-19} C and produces a radius of 0.0183 m in the magnetic field?

$$m = \frac{qr^2B^2}{2\Delta V} = \frac{(1.6\ x\ 10^{-19}C)(0.0183\ m)^2(2T)^2}{2(1{,}000V)} = 1.07\ x\ 10^{-25}kg$$ This is a zinc atom!

Velocity selector – When a charged particle moves through an electric field it experiences a force along the field, this electric force does not depend on its velocity. When the charged particle travels through a magnetic field it experiences a force perpendicular to the field and the magnetic force does depend on the velocity of the charged particle. There is a specific velocity that a charge particle will experience equal but opposite force from the electric and magnetic fields therefore will have no net force acting. This happens at only one velocity and any other velocities will be deflected.



At the exit screen:

- Point P the velocity is too large creating a larger magnetic force than electric force.
- Point Q the velocity is too small creating a smaller magnetic force than electric force.
- The charged particle only goes through the aperture when the electric force is equal to the magnetic force.

If the electric field is pointing up and the magnetic field is out of the page, then when a charged particle enters the fields.

$$\left|\vec{F}_E\right| = \left|\vec{F}_M\right|$$

$$q|\vec{\mathbf{E}}| = q|\vec{v}||\vec{\mathbf{B}}|$$

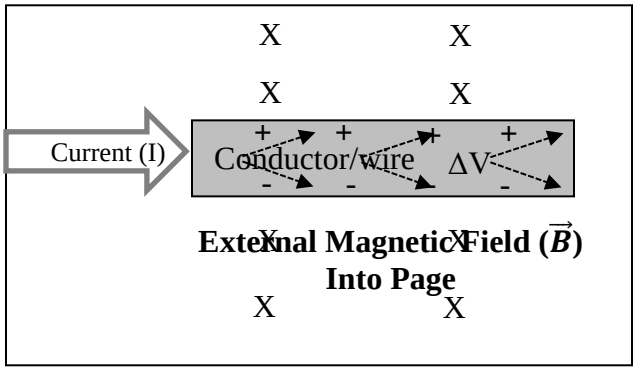
Therefore, the only velocity that will not have a net force and not be defected is

$$|\vec{\mathbf{v}}| = \frac{|\vec{\mathbf{E}}|}{|\vec{\mathbf{B}}|}$$

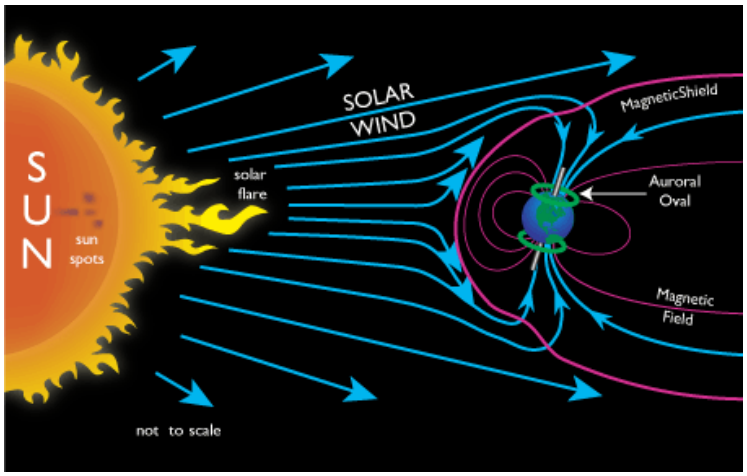
Example: Determine the velocity a charged particle would have pass straight through a velocity sector with an electric field of 1,200 V/m and a magnetic field of 2 x 10^{-3} T?

$$v = \frac{E}{B} = \frac{1200\ V/m}{2\ x\ 10^{-3} T} = 6\ x\ 10^{5}\ m/s$$

Hall effect – When a stationary, current carrying, conductor is in a perpendicular uniform magnetic field, the charge is separated by the magnetic force due to the charges motion in the field. The negative charge moves to one side and the positive to the other creating a potential difference across the short side of the conductor. This effect is used to measure the strength of the magnetic field.



Earth's aurora borealis or Northern Lights– The earth magnetic field interacts with charged particles coming from the sun (solar wind) these charged particles are caught in the magnetic field of the earth and move towards the north and south poles, as the charged particles interact with the atmosphere closer to the earth, they emit light that is seen as the aurora borealis



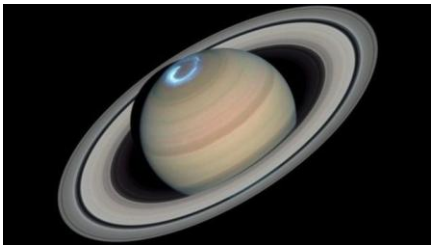


This same effect can be seen on other planets with magnetic fields.

Jupiter



Saturn



<u>Bubble Chambers</u> are used to indentify sub atomic particle that are formed when very high energy particles collide. The traks that are shown are the product of the coldiding particles in a strong magnetic field. The directions, radius and shape of the lines can be used to identify the parteicles that are created and what they decay into.

